

PROBLEM STATEMENT #18 | Healthcare & Telemedicine

Clinical Trial Subject Recruitment Engine

1. Problem Context & Overview

Medical research institutions require an eligibility pre-screening platform for clinical trials, tracking subject inclusion/exclusion criteria, informed consent milestones, and visit compliance logs.

Target Stakeholders / Actors: *Trial Candidate, Clinical Researcher*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall evaluate candidate survey responses against trial protocol inclusion/exclusion criteria and classify eligibility status.

Sample Acceptance Criteria: Pass: Ineligible candidate flagged with specific failed criteria; Fail: Subject with exclusion criteria marked eligible.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: The recruitment database must comply with 21 CFR Part 11 electronic record standards including cryptographic audit trails.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.